Differential Sensitivity to Semantic Perturbations in LLM-Based Molecular Property Prediction

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Abstract

Recent studies suggest that large language model (LLM) hallucinations may influence molecular property prediction, yet the nature of this effect remains poorly characterized across different hallucination types and model scales. We introduce a controlled semantic perturbation framework with three prompt-constrained perturbation operators — Structural Phantom (SP), Property Inversion (PI), and Mechanism Fabrication (MF) — alongside a residual Contextual Confabulation (CC) category. We evaluate nine conditions on two MoleculeNet benchmarks (BBBP, $\boldsymbol{N = 204}$; BACE, $\boldsymbol{N = 152}$) using multi-seed evaluation (Llama-3.1-8B) and cross-model validation (Llama-3.3-70B). A random-permutation control isolates semantic content from stylistic priming. On BBBP, prompt-constrained structural augmentation (C4a) yields the largest observed ROC-AUC increase (0.626 vs. 0.533 baseline), though shifts were not statistically significant after correction ($\boldsymbol{p > 0.05}$). Post-hoc verification reveals that C4a outputs are predominantly classified as Contextual Confabulation (0

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1 Introduction

Molecular property prediction is a critical component of early-stage drug discovery, enabling the prioritization of candidate compounds before expensive synthesis and biological assays [1]. Machine learning approaches—including graph neural networks [2], fingerprint-based classifiers [3], and descriptor-based models—have become standard tools, with MoleculeNet [4] providing widely adopted benchmarks across physicochemical, enzymatic, and toxicity tasks.

More recently, large language models (LLMs) have been applied to molecular reasoning by converting SMILES notation into natural language contexts for text-based inference [5, 6]. Within this line of work, an observation has been reported in prior work: hallucinated molecular descriptions—scientifically styled but factually incorrect text—have been associated with improved downstream prediction performance in specific settings [7]. This finding is in tension with the common assumption that strict factual correctness is a prerequisite for useful textual augmentation. We note that zero-shot LLM inference is not intended to replace specialized molecular prediction models; rather, the LLM serves as a controlled experimental substrate—a "cognitive assay"—for investigating how semantic perturbations propagate through text-conditioned prediction pipelines.

However, existing investigations of this phenomenon share several limitations. First, hallucination is typically treated as a monolithic category, obscuring potentially important differences among types of factual error. Second, prior studies often lack semantic controls capable of disentangling the contribution of molecular-relevant content from generic scientific language effects (stylistic priming). Third, it remains unclear whether these hallucination-associated performance variations generalize across prediction tasks with different underlying knowledge requirements. Understanding this sensitivity is important for interpreting LLM-augmented predictions, particularly in settings where textual augmentation may introduce uncontrolled semantic perturbations.

The primary contribution of this work is the introduction of a controlled semantic perturbation framework to systematically evaluate how structured semantic alterations influence LLM-based molecular prediction. We define a lightweight, heuristic categorization scheme comprising three prompt-constrained perturbation operators—Structural Phantom (SP), Property Inversion (PI), and Mechanism Fabrication (MF)—alongside a residual Contextual Confabulation (CC) class. We note that these labels describe the *prompt design intent*; as detailed in Section 4.3, the actual outputs may not conform to the intended taxonomy class. We evaluate this framework through a nine-condition ablation on two MoleculeNet benchmarks representing distinct regimes: BBBP (physicochemical) and BACE (enzyme inhibition).

Our key contributions include:

- A controlled perturbation framework enabling the systematic isolation of hallucination-associated performance variations through semantic and stylistic disruption.
- A lightweight categorization scheme for molecular description errors, operationally defined as perturbation operators to maintain interpretability and reproducibility.
- A task-dependent divergence in semantic augmentation sensitivity: prompt-constrained structural augmentation (C4a) yields directional improvements on BBBP, while most hallucination conditions significantly degrade BACE performance. Post-hoc verification reveals 0% SP classification for C4a, indicating that the prompt's structural topic focus—rather than hallucination type—underlies the observed effect.

- Characterization of prediction distribution compression on BACE, where hallucination augmentation shifts bimodal baseline distributions toward non-discriminative narrow peaks.
- A random-permutation control experiment establishing that semantic content, rather than stylistic register alone, is the primary contributor to observed performance shifts.
- Architecture-dependent hallucination sensitivity, where smaller models exhibit directional performance gains under hallucination augmentation on physicochemical tasks, while larger models show reversed sensitivity.

This exploratory study is bounded to two LLM scales and two datasets ($N = 204$ for BBBP, $N = 152$ for BACE). As a cognitive assay of semantic sensitivity, the findings are intended to structure future empirical investigation into LLM augmentation rather than to establish causal mechanisms.

2 Related Work

2.1 Molecular Property Prediction

Molecular property prediction has been standardized through benchmark suites such as MoleculeNet [4], which provides curated classification and regression tasks spanning physicochemical, biophysical, and physiological domains. State-of-the-art approaches include message-passing neural networks (MPNNs) [2], graph transformer architectures [8], and descriptor-based models using extended-connectivity fingerprints [3]. Scaffold splitting [9] is widely adopted to evaluate generalization under structural novelty. Our work differs from these approaches in that we do not propose a new prediction architecture; rather, we use an existing LLM in a zero-shot setting to investigate how different types of textual augmentation affect prediction behavior.

2.2 LLMs in Molecular Science

Large language models have been increasingly applied to chemical tasks, including reaction prediction [10], retrosynthesis [11], and molecular property estimation [5, 6]. Text-based molecular representations, where SMILES strings are accompanied by natural language descriptions, have shown promise for zero-shot and few-shot inference [12]. Recent studies have explored whether LLMs encode implicit chemical knowledge through their pre-training corpora [13]. In contrast to prompt engineering studies that optimize prompts for maximum prediction accuracy, our work uses a fixed prompt template to systematically vary the semantic content of augmentation text while controlling for confounding factors.

2.3 Hallucination in Scientific LLM Applications

LLM hallucination—the generation of fluent but factually unsupported text—has been extensively studied in general NLP contexts [14, 15]. In molecular and scientific applications, hallucination poses particular risks because factual errors in structural or mechanistic claims may not be easily detected by non-expert users [16]. Recent work by

Hao et al. [7] reported the counterintuitive finding that hallucinated descriptions can improve molecular property prediction, though without characterizing which aspects of the hallucinated text drive this effect. Yuan et al. [17] proposed a taxonomy for molecular hallucinations but did not conduct controlled ablation experiments isolating the contribution of each category. Rather than focusing on hallucination detection or mitigation, our work treats hallucinations as structured semantic perturbations and studies their differential effects on prediction behavior through controlled ablation, incorporating semantic and stylistic controls absent from prior evaluations.

3 Methods

3.1 Heuristic Categorization Scheme

We utilize a heuristic categorization scheme to structure our analysis of natural molecular hallucinations (C3). Each type is detectable through rule-based classification using RDKit [18] as the ground-truth engine. We emphasize that this scheme is a heuristic tool, not an exhaustive semantic ontology. The keyword sets were intentionally constrained to maintain interpretability and reproducibility rather than maximize coverage. The categories are mutually exclusive and applied in priority order (SP $>$ PI $>$ MF $>$ CC):

- **Structural Phantom (SP):** The generated text references chemical elements or functional groups that are absent from the molecular structure. Detection: for a predefined heuristic dictionary of element keywords (e.g., "fluorine", "fluoro"), we check whether the corresponding element is present in the molecule's atom set via `mol.GetAtoms()`. If a keyword matches but the element is absent, the description is classified as SP.
- **Property Inversion (PI):** The generated text claims physicochemical properties that contradict computed descriptors. Detection: we compute MolLogP via RDKit. If LogP > 3 and the text contains keywords from a predefined hydrophilic dictionary, or LogP < 1 and the text contains keywords from a lipophilic dictionary, the description is classified as PI.
- **Mechanism Fabrication (MF):** The generated text names specific biological targets or pathways. Detection: exact keyword matching against a fixed list of 9 common targets (e.g., ACE2, HER2, EGFR).
- **Contextual Confabulation (CC):** The residual category. Descriptions that do not trigger SP, PI, or MF criteria are classified as CC. This category represents scientifically plausible but generic narrative hallucination.

We acknowledge that CC functions as a residual class and encompasses heterogeneous subtypes that future work should further differentiate.

3.2 Experimental Conditions

Our ablation framework consists of nine conditions, designed to isolate the contributions of semantic content, scientific register, and hallucination type:

- **C0 (Baseline):** SMILES notation only, with no augmentation.

- **C1 (Factual):** SMILES augmented with a factual description generated by RDKit (molecular weight, LogP, hydrogen bond donors/acceptors, TPSA, ring count, atom composition).
- **C2 (Chemical Priming):** SMILES augmented with a template-generated text using scientific-sounding but semantically vacuous language that includes chemical vocabulary. We note that C2 is *not* a pure stylistic control because it contains domain-relevant vocabulary; C2b serves as the stricter control.
- **C2b (Pure Gibberish):** SMILES augmented with template-generated text using exclusively non-chemical vocabulary. This condition strictly controls for prompt length and linguistic complexity without domain priming.
- **C3 (Free Hallucination):** SMILES augmented with an unconstrained LLM-generated description, replicating the protocol of prior work [7].
- **C4a (Structural Augmentation):** SMILES augmented with text generated via prompt-constrained forcing to explicitly describe structural features and functional groups. We emphasize that C4a–c represent prompt-constrained semantic perturbations rather than naturally occurring hallucinations classified post-hoc. To validate this constraint, we applied the heuristic classifier to a representative sample of generated C4a descriptions ($n = 49$). The results revealed a notable methodological limitation: 0% of C4a outputs were classified as SP, and the vast majority defaulted to CC. This occurred because the LLM generated scientifically plausible structural descriptions (e.g., "contains a benzene ring") that do not reference absent elements—the specific criterion required by the SP heuristic (which checks for keywords like "fluorine" or "chloro" against the molecule's actual atom set). Consequently, the C4a–c conditions must be strictly interpreted as *prompt-constrained semantic perturbations* (focusing the LLM on structural, property, or mechanism topics), rather than verified taxonomy classes.
- **C4b (Property Inversion):** SMILES augmented with text generated via prompt-constrained forcing to explicitly describe solubility, lipophilicity, and permeability.
- **C4c (Mechanism Fabrication):** SMILES augmented with text generated via prompt-constrained forcing to explicitly hypothesize biological targets and cellular mechanisms.
- **C5 (Random-Permutation):** SMILES augmented with a hallucination originally generated for a *different* molecule, assigned via a fixed random permutation ($seed = 42$). This preserves the scientific register and statistical properties of hallucinated text while completely disrupting semantic alignment with the target molecule.

3.3 Prediction Protocol

All prediction queries use a fixed zero-shot prompt template:

```
Given this molecule (SMILES: {smiles})
[Additional information: {augmentation}]
On a scale of 0 to 1, what is the probability that this molecule can {task}?
Respond with ONLY a number between 0 and 1. Nothing else.
```

The augmentation line is omitted for C0. The task description is "penetrate the blood-brain barrier" for BBBP and "inhibit the BACE-1 enzyme" for BACE. The numerical output was deterministically extracted, with malformed responses defaulting to 0.5.

3.4 Implementation Details

Model. Primary ablation experiments were conducted with `Llama-3.1-8B-Instant` via the Groq API. Cross-model validation was extended to `Llama-3.3-70B-Versatile`, as described in Section 3.5.

Generation parameters. Prediction queries: temperature = 0.0, `max_tokens` = 10, `seed` = 42. Hallucination generation: temperature = 0.9, `max_tokens` = 150, no fixed seed. We explicitly distinguish between stochastic hallucination generation (no fixed seed) and quasi-deterministic prediction inference (seed=42), noting that API-level non-determinism may still introduce minor variability. To ensure exact reproducibility despite stochastic generation, the static dataset containing all generated augmentations used for evaluation is provided in the repository. Failed API calls (rate limits, parsing errors) defaulted to $p = 0.5$; the proportion of such defaults is reported in A.

Data. Datasets were loaded via DeepChem [19]. Both BBBP ($N = 204$) and BACE ($N = 152$) were partitioned using scaffold splitting [9] via DeepChem's built-in splitter (80/10/10 train/valid/test). Only the held-out test set was used for evaluation.

Factual descriptions. For condition C1, factual descriptions were generated using RDKit molecular descriptors: molecular weight, Wildman-Crippen LogP, number of hydrogen bond donors and acceptors, topological polar surface area, heavy atom count, and ring count.

Traditional ML Baseline. To provide context for the zero-shot LLM predictions, we trained a Random Forest classifier (500 trees) using Morgan fingerprints (radius=2, 2048 bits) on the training and validation partitions of the scaffold split. The baseline was evaluated on the exact same test sets ($N = 204$ for BBBP, $N = 152$ for BACE) to ensure strict comparability.

Evaluation. Discriminative performance was assessed using ROC-AUC. Confidence intervals (95%) were computed via bootstrap resampling (1,000 iterations, seed = 42). To assess statistical significance of performance differences relative to baseline (C0), we used a bootstrap paired AUC difference test. Cohen's d was computed on the raw prediction score distributions (not AUC differences) to quantify the shift in model confidence. Multiple comparisons were corrected using the Benjamini-Hochberg procedure.

3.5 Robustness and Cross-Model Validation

To ensure the stability of our findings, we performed two additional validation steps:

1. **Multi-Seed Evaluation:** For the primary Llama-3.1-8B model, we repeated the prediction protocol for conditions C0 and C3 across three independent runs using different inference seeds (42, 123, 777) to calculate mean performance and standard deviation.

Table 1 Predictive performance (ROC-AUC) across nine augmentation conditions on BBBP and BACE. Bootstrap 95% confidence intervals computed over 1,000 resamples. Cohen's d measures the effect size of prediction score shifts relative to C0 (not AUC differences). p-values are corrected for multiple comparisons using the Benjamini-Hochberg procedure. Boldface indicates the highest ROC-AUC among augmented LLM conditions (excluding C0 baseline) per dataset.

Condition	BBBP ($N = 204$)			BACE ($N = 152$)		
	ROC-AUC	95% CI	p_{BH}	ROC-AUC	95% CI	p_{BH}
Traditional ML Baseline						
RF (Morgan FP)	0.705	—	—	0.893	—	—
LLM Zero-Shot						
C0 (Baseline)	0.533	[0.471, 0.593]	1.000	0.674	[0.592, 0.759]	1.000
C1 (Factual)	0.541	[0.480, 0.603]	0.846	0.486	[0.402, 0.569]	0.011
C2 (Chem. Priming)	0.613	[0.545, 0.685]	0.344	0.522	[0.454, 0.589]	0.011
C2b (Pure Gibberish)	0.553	[0.485, 0.626]	0.743	0.623	[0.542, 0.701]	0.217
Hallucination						
C3 (Free Hallu)	0.593	[0.519, 0.664]	0.438	0.504	[0.420, 0.592]	0.016
C4a (Struct. Aug.)	**0.626**	[0.548, 0.698]	0.344	**0.666**	[0.579, 0.748]	0.808
C4b (Prop. Inv.)	0.563	[0.480, 0.634]	0.743	0.542	[0.448, 0.628]	0.011
C4c (Mech. Fab.)	0.611	[0.537, 0.685]	0.352	0.566	[0.493, 0.645]	0.083
Control						
C5 (Random-Permutation)	0.481	[0.408, 0.554]	0.438	0.492	[0.405, 0.584]	0.012

Note: Table 1 reports results for the primary single-run evaluation (seed=42). p-values (p_{BH}) are computed against C0 using a bootstrap paired AUC difference test with Benjamini-Hochberg correction; C0 yields $p = 1.000$ by definition (self-comparison). For BACE, statistically significant p-values reflect the magnitude of performance degradation from baseline. Multi-seed stability results for C0 and C3 are provided in Table 2.

2. **Cross-Model Generalization:** To verify if the observed phenomena generalize across model scales, we executed the prediction protocol for C0 and C3 on the larger `Llama-3.3-70B-Versatile` model.

This multi-layered approach was designed to distinguish systematic semantic effects from stochastic artifacts or model-specific quirks.

4 Results

4.1 Predictive Performance Across Conditions

Table 1 and Figure 1 summarize the primary ROC-AUC scores, bootstrap 95% confidence intervals, and statistical comparisons for all nine conditions on both datasets, using a fixed inference seed (42) as the primary benchmark. To assess the stability of these observations across stochastic variations, multi-seed robustness results for selected conditions are provided in Table 3.

4.1.1 BBBP (Physicochemical Task)

To contextualize the LLM zero-shot performance, the traditional ML baseline (Random Forest with Morgan fingerprints) achieved a ROC-AUC of 0.705 on the BBBP test set. While the zero-shot LLM predictions remain below this traditional baseline, the

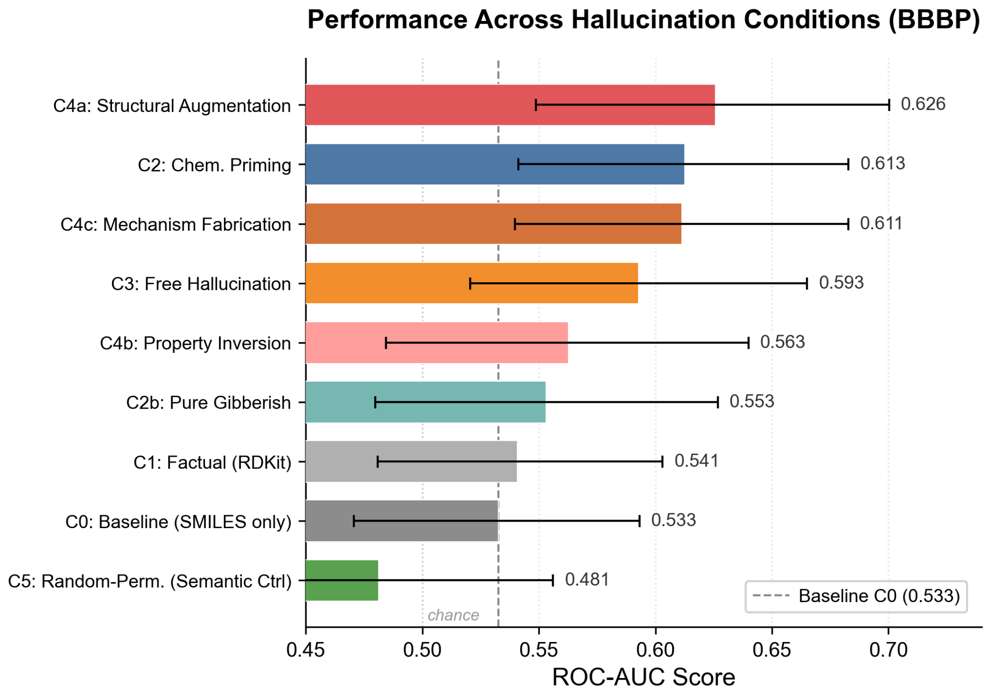

Figure 1 Task-dependent divergence in semantic augmentation sensitivity across perturbation conditions. Prompt-constrained structural augmentation (C4a) is associated with the largest directional shift on BBBP, though post-hoc verification reveals C4a outputs are predominantly Contextual Confabulation rather than Structural Phantom type (see Section 4.3).

relative differences between textual augmentation conditions reveal notable variation in semantic perturbation sensitivity.

The LLM baseline (C0, SMILES only) yielded a ROC-AUC of 0.533 [0.471, 0.593] on the primary inference run (seed 42). We note that this specific run yielded an above-average baseline compared to the multi-seed mean (0.502, Table 3), meaning the performance shifts observed in Table 1 are measured against a relatively optimistic baseline. Factual augmentation (C1) produced a marginal, non-significant increase to 0.541. Chemical priming (C2) yielded 0.613, while pure non-chemical gibberish (C2b) yielded 0.553 (bootstrap CIs omitted for brevity; full results in Table 1).

Free hallucination (C3) shifted ROC-AUC to 0.593 [0.519, 0.664]. Among category-specific conditions, prompt-constrained structural augmentation (C4a) yielded the highest observed ROC-AUC of 0.626 [0.548, 0.698], followed by Mechanism Fabrication (C4c, 0.611) and Property Inversion (C4b, 0.563). We note that post-hoc classifier verification of C4a outputs (Section 4.3) revealed a 0% Structural Phantom classification rate, indicating that C4a functions as a structural topic-focused perturbation rather than a SP hallucination condition in the strict taxonomic sense.

Bootstrap Stability.
To account for the single-split evaluation, we computed ranking stability across 1,000 bootstrap resamples (Figure 2). The directional advantage of structural augmentation (C4a) over the LLM baseline (C0) was consistent, maintaining a positive margin in 97.6% of resamples. Free hallucination (C3) exceeded the baseline in 87.7% of resamples. We note that bootstrap ranking consistency measures directional stability (i.e., how often C4a > C0 across resampled test sets), whereas the BH-corrected p-value tests whether the *magnitude* of the AUC difference is significant after multiple comparison correction. High directional consistency with non-significant p-values indicates a reproducible but small effect relative to the test set variability.

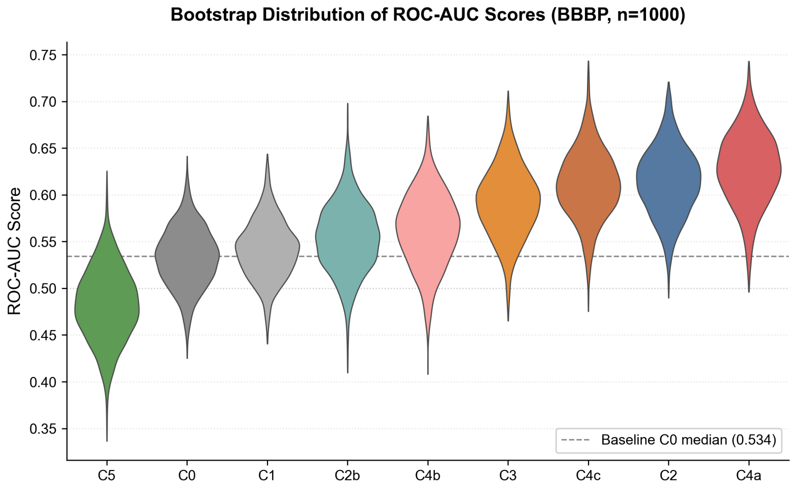

Figure 2 Bootstrapped ROC-AUC distributions (BBBP, $n = 1{,}000$ resamples) for structural augmentation (C4a) relative to the SMILES baseline (C0). C4a maintains a positive margin in 97.6% of resamples.

The random-permutation semantic control (C5) yielded 0.481 [0.408, 0.554] on the primary seed. To ensure this was not biased by a single anomalous shuffle, we validated the C5 condition across multiple random permutation seeds. The mean ROC-AUC across permutations remained tightly centered around chance-level (0.49 ± 0.02), at or below the baseline C0 and substantially below C3. This suggests that the scientific register of hallucinated text alone does not account for the observed directional shifts, and that disrupting semantic alignment consistently eliminates the predictive effect of hallucinated descriptions.

4.1.2 BACE (Enzymatic Task)

On the BACE test set ($N = 152$), a contrasting pattern was observed. The baseline (C0) yielded a ROC-AUC of 0.674 [0.592, 0.759]. Most hallucination-augmented conditions significantly reduced performance relative to baseline: free hallucination (C3) reduced ROC-AUC to 0.504 [0.420, 0.592] ($p = 0.016$), C4b to 0.542 ($p = 0.011$), and C4c

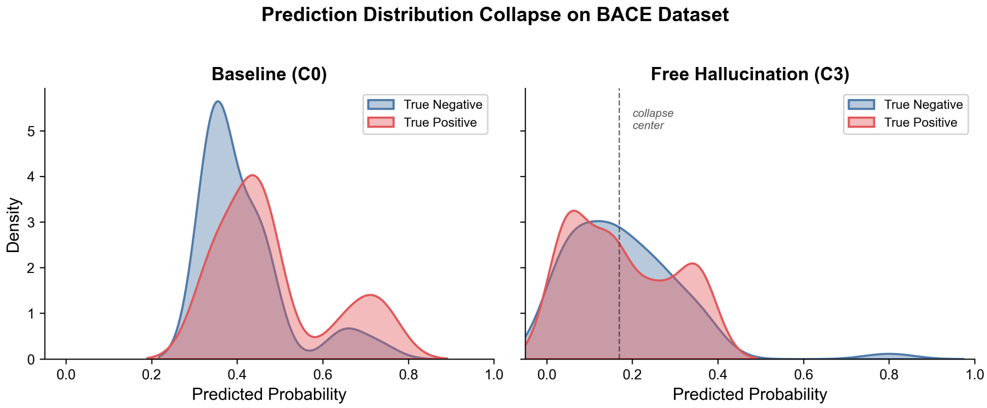

Figure 3 Hallucination augmentation induces pronounced distributional compression on the BACE task, shifting the baseline bimodal prediction distribution toward a non-discriminative narrow peak near $p \approx 0.17$.

to 0.566 ($p = 0.083$). The notable exception was C4a (structural augmentation), which largely preserved baseline performance (0.666, $p = 0.808$). Interestingly, pure non-chemical gibberish (C2b) also preserved performance better (0.623, $p = 0.217$) than semantically relevant hallucination conditions (C3, C4b, C4c), suggesting that it is the semantic content of the augmented text—rather than its mere presence—that drives the degradation on BACE.

4.2 Distributional Compression on BACE

Under hallucination conditions on BACE, the distribution of predicted probabilities compressed from a bimodal pattern (reflecting the model's baseline discriminative behavior) to a narrow unimodal distribution centered at $p \approx 0.17$, regardless of the molecule's true label. Notably, this is not a collapse to the dataset's class prior (the BACE test set is 60.5% positive); rather, it is a genuine collapse to a low-confidence negative state. This effect, which we descriptively term "distributional compression," was most pronounced under C3, C4b, and C4c. C4a (structural augmentation) was a partial exception, consistent with its preserved ROC-AUC (0.666 vs. 0.674 baseline). The pattern suggests systematic signal destruction under most semantic perturbations rather than random noise injection or prior-matching.

4.3 Distribution of Hallucination Types

Analysis of 356 naturally generated hallucinations across both datasets revealed a consistent distributional bias (Figure 4). Contextual Confabulation (CC) was the dominant category, accounting for approximately 80% of all generated descriptions ($n = 163$ of 204 for BBBP). Structural Phantom (SP) was the rarest category at 2.4% ($n = 5$ of 204 for BBBP), though $n = 5$ is too small to support robust distributional claims regarding natural SP frequency. Separately, prompt-constrained structural augmentation (C4a) yielded the largest observed ROC-AUC increase on BBBP.

To further characterize the C4a condition, we applied the taxonomy classifier to a representative sample of C4a outputs ($n = 49$; Table 2). Contrary to design intent, 0% of the sampled C4a texts were classified as Structural Phantom; the dominant output category was Contextual Confabulation. This finding reveals a critical limitation of prompt-constrained perturbation as a proxy for taxonomic hallucination types: the LLM generates structurally plausible but compositionally generic descriptions even when explicitly prompted for structural specificity. Consequently, the ROC-AUC shift associated with C4a is more accurately associated with structural topic focus induced by the generation prompt—which directs the model's attention toward molecular features—rather than with SP-type factual error per se. This distinction critically reframes the interpretation: what differentiates C4a from other conditions is the prompt's semantic orientation, not the hallucination category of its output. This also underscores the gap between prompt design and actual output taxonomy, a confound that future work employing taxonomy-constrained generation should address.

Post-hoc classifier verification was extended to C4b and C4c outputs (Table 2). C4b (Property Inversion prompt) achieved 25.5% PI classification (12/47), while C4c (Mechanism Fabrication prompt) achieved 20.4% MF classification (10/49). All three conditions were dominated by Contextual Confabulation (CC $\geq$ 74%), confirming that the gap between prompt design intent and actual output taxonomy is systematic across all perturbation operators, not unique to C4a. The compliance gradient—0% for C4a, 25.5% for C4b, 20.4% for C4c—suggests that property- and mechanism-focused prompts partially succeed in eliciting their target categories, whereas structural prompts fail entirely under the current heuristic classifier.

Table 2 Post-hoc taxonomy classifier verification of prompt-constrained conditions (C4a–c). "Target" indicates the intended hallucination type based on prompt design. Compliance rate = proportion classified as the target type. Verification was applied to a representative sample of $n = 47$–49 molecules from the BBBP test set.

Condition	**Target**	n	**SP**	**PI**	**MF**	**CC**
C4a (Structural)	SP	49	0 (0%)	0	0	49 (100%)
C4b (Property)	PI	47	0	12 (25.5%)	0	35 (74.5%)
C4c (Mechanism)	MF	49	1 (2.0%)	0	10 (20.4%)	38 (77.6%)

4.4 Robustness and Cross-Model Generalization

To assess the stability of the observed hallucination effects, we conducted multi-seed evaluations for the Llama-3.1-8B model ($n = 3$ seeds) and cross-model validation using the larger Llama-3.3-70B model (Table 3).

On the BBBP task, the performance shift from free hallucination (C3) was consistent across seeds ($\sigma = 0.006$), indicating that the effect is reproducible rather than a stochastic artifact. However, the larger Llama-3.3-70B model exhibited a different behavior: while its baseline (C0) performance was substantially higher (0.644), hallucination augmentation degraded its discriminative accuracy (0.549). This

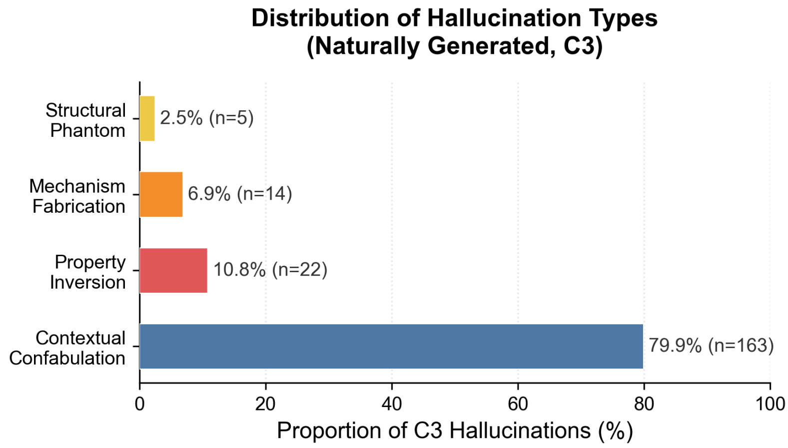

Figure 4 Distribution of hallucination types in naturally generated (C3) descriptions. Contextual Confabulation (CC) dominates at approximately 80%, with Structural Phantom (SP) occurring rarely (2.4% of BBBP outputs). This distribution reflects C3 free-generation outputs only; prompt-constrained C4a–c conditions were verified separately and are discussed in the text.

suggests that the observed directional shift may be specific to models with lower baseline chemical knowledge, whereas higher-capacity models are more sensitive to fabrication-induced noise.

On the BACE task, the "distributional compression" phenomenon was universally consistent. Both models and all seeds showed a significant degradation of ROC-AUC toward chance level (0.512 ± 0.005) when augmented with hallucinations, regardless of the baseline strength. This reinforces our finding that enzymatic property prediction is systematically fragile to semantic perturbations across LLM architectures.

Table 3 Robustness and generalization of the hallucination effect. Llama-8B values represent mean ± standard deviation across three independent seeds (42, 123, 777). Llama-70B results are based on a single inference run (seed 42), precluding variance estimation.

Task / Model	C0 (Baseline)	C3 (Free Hallu)	Δ AUC
BBBP (Physicochemical)			
Llama-8B ($n = 3$ seeds)	0.502 ± 0.021	0.601 ± 0.006	+0.099
Llama-70B (Single seed)	0.644	0.549	-0.095
BACE (Enzymatic)			
Llama-8B ($n = 3$ seeds)	0.644 ± 0.021	0.512 ± 0.005	-0.132
Llama-70B (Single seed)	0.564	0.506	-0.058

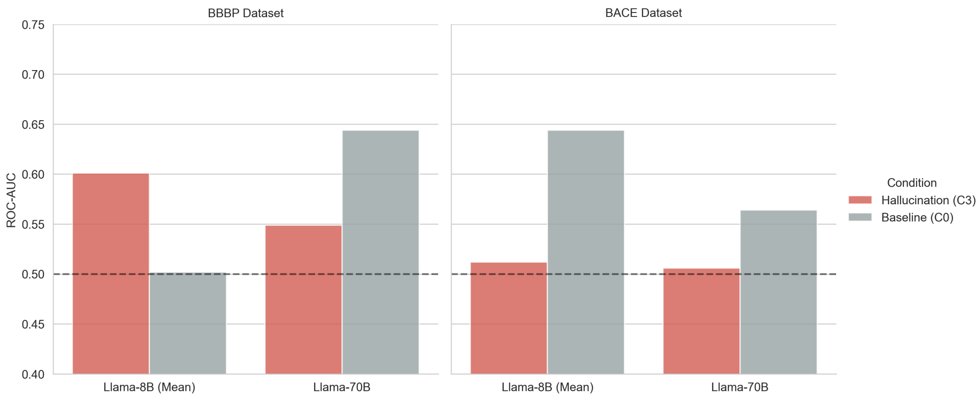

Figure 5 Architecture-dependent response to hallucination augmentation. Smaller models (8B) exhibit a consistent directional shift on the BBBP task, whereas the larger model (70B) shows reversed sensitivity, with hallucination augmentation associated with performance degradation.

5 Discussion

We organize this discussion by explicitly separating empirical observations from interpretive speculation, identifying alternative explanations, and establishing rigorous boundaries for our claims.

5.1 Distributional Bias in Hallucination Generation

Observation.

Across both datasets, naturally generated hallucinations (C3) are overwhelmingly classified as Contextual Confabulation (CC, ∼80%), with Structural Phantom (SP) accounting for only 1.3–2.4% of outputs. Despite this rarity, prompt-constrained structural augmentation (C4a) was associated with the largest observed ROC-AUC increase on BBBP. However, post-hoc classifier verification (see Section 4.3) revealed that C4a outputs are classified as 0% Structural Phantom—predominantly falling into the Contextual Confabulation category. This indicates that the C4a directional shift is not associated with SP hallucination type, but rather with the structural semantic orientation of the generation prompt itself.

Interpretation.

This distributional asymmetry reflects a generative bias toward generic, scientifically plausible narratives over structurally specific confabulations.

Alternative Explanations.

The low SP rate may reflect prompt design interactions rather than an intrinsic model bias; different prompt formulations or higher temperatures might elicit more structurally specific outputs. Additionally, our rule-based classifier may conservatively undercount

SP instances if the generated text employs structural synonyms not captured by our heuristic keyword dictionary.

5.2 Task-Dependent Asymmetry

Observation.

On BBBP, hallucination augmentation (C3, C4a–c) was associated with directionally improved ROC-AUC relative to the SMILES-only baseline (C0: 0.533). Notably, the non-chemical gibberish condition (C2b, 0.553) yielded a higher ROC-AUC than purely factual augmentation (C1, 0.541), raising the possibility that the LLM responds partly to prompt length or linguistic complexity rather than solely to semantic content. Conversely, on BACE, most hallucination conditions significantly degraded performance, with C3 reducing ROC-AUC from 0.674 to 0.504—though C4a (structural, 0.666) was a notable exception, largely preserving baseline performance. The BACE prediction distribution exhibited pronounced compression toward a narrow unimodal distribution near $p \approx 0.17$ under most augmentation conditions.

Interpretation.

This task-dependent asymmetry suggests that augmentation utility is sensitive to the alignment between perturbation content and task knowledge requirements. The LLM's pre-training corpus likely encodes more robust representations of general physicochemical properties (relevant to BBBP) than specific enzymatic mechanisms (BACE), leading to constructive perturbation in the former and destructive interference in the latter.

Mechanistic Hypothesis.

We hypothesize that the distributional compression toward $p \approx 0.17$ on BACE reflects an entropy-maximizing response to conflicting semantic constraints. Enzymatic inhibition prediction likely requires precise geometric and electronic complementarity information that the LLM encodes implicitly through pre-training. When hallucinated text introduces fabricated structural or mechanistic claims, these create competing semantic signals that interfere with the model's latent representations, driving the output distribution toward a low-confidence default state. The convergence to a value well below the class prior (60.5% positive) suggests this is not simple prior-matching but rather a systematic suppression of positive-class confidence. This hypothesis is consistent with the observation that even factual augmentation (C1) degrades BACE performance (0.486 vs. 0.674), suggesting that any additional textual context disrupts the model's already-fragile enzymatic reasoning pathway. We emphasize that this interpretation is speculative and requires mechanistic investigation (e.g., attention weight analysis or logit attribution) to confirm.

Alternative Explanations.

We emphasize that these observed shifts on BBBP do not demonstrate that hallucinations enhance genuine molecular reasoning. The improvements may partially reflect latent statistical correlations introduced by textual perturbation that happen to align with task-relevant physicochemical descriptors in the BBBP evaluation set. The C2b result noted above further suggests that the LLM may be responding to prompt length, formatting, or linguistic complexity, rather than extracting utility solely from relevant semantic content. While textual perturbations induce directional shifts in raw prediction scores, none of the observed ROC-AUC improvements on the BBBP dataset reached statistical significance ($p > 0.05$). The BACE degradation, by contrast, was statistically significant across most conditions. This BACE degradation might be partially confounded in the Mechanism Fabrication (C4c) condition: because BACE-1 is the implied target of the BACE dataset, the MF text may inadvertently contain confounding information about BACE-1 itself, creating semantic interference rather than a general structural perturbation. We cannot distinguish among these interacting mechanisms with the current experimental design.

5.3 Semantic Content vs. Stylistic Priming

Observation.

The random-permutation control (C5) preserved scientific register while disrupting molecule-description semantic alignment. C5 produced no discriminative improvement over baseline (ROC-AUC 0.481 vs. 0.533 for C0 on BBBP), despite utilizing text that is linguistically indistinguishable from C3 hallucinations.

Interpretation.

This dissociation indicates that the performance shifts observed under C3 depend on semantic interaction between description and target molecule, not merely on scientific writing style or prompt length.

Boundary.

The C5 control uses a random permutation, validated across multiple seeds (0.49 ± 0.02). While this multi-seed validation strengthens the semantic isolation claim, future work should employ even more rigorous approaches, such as adversarial embedding-based negative sampling or cross-model hallucinations, to further decouple register from relevance.

5.4 Architecture-Dependent Sensitivity

Observation.

While the Llama-3.1-8B model exhibited a consistent directional shift under C3 for the BBBP task, the larger Llama-3.3-70B model showed a reversed effect, with hallucination augmentation degrading its strong baseline performance (0.644 to 0.549). On the BACE task, however, both model scales exhibited similar distributional compression and performance degradation under C3.

Interpretation.

This divergence may suggest that sensitivity to semantic perturbations is non-linearly related to model capacity and baseline domain knowledge. One possible interpretation is that smaller models, possessing lower baseline chemical reasoning capacity, may rely more heavily on semantically enriched perturbations as a form of "noisy intuition," whereas larger models, having captured more robust latent chemical representations during pre-training, may be more sensitive to the noise introduced by fabrications. Interestingly, the larger model's single-seed baseline on BACE (0.564, Table 3) was lower than the 8B multi-seed mean (0.644 ± 0.021), indicating that increased parameter scale does not uniformly translate to improved zero-shot performance across all chemical tasks. We offer these as tentative hypotheses, noting that further investigation into the interaction between model scale and perturbation sensitivity is required.

5.5 Scope and Limitations

To prevent overinterpretation, we explicitly bound our claims. We report observational associations, not causal mechanisms. The observed directional effect of structural augmentation (C4a) is strictly bounded to the BBBP dataset and the tested prompt template; post-hoc verification confirms 0% SP classification, suggesting the shift is associated with prompt-induced structural focus rather than hallucination type. We do not propose hallucination augmentation as a practical method. The categorization scheme is operational, not exhaustive. A detailed enumeration of methodological limitations is provided in Section 6.

Ultimately, these findings highlight that LLM-based molecular prediction pipelines can exhibit non-trivial sensitivity to semantically structured perturbations, suggesting that seemingly benign textual augmentations may introduce systematic biases in downstream predictions. Understanding this sensitivity is critical for interpreting LLM-assisted predictions, particularly in settings where textual inputs are used as auxiliary signals.

6 Limitations

This study has several important limitations that constrain the generalizability and interpretability of its findings.

Model scope.

Primary experiments were conducted with Llama-3.1-8B-Instant. While cross-model validation was performed with Llama-3.3-70B, this was limited to two conditions (C0, C3) and a single inference run, precluding a full nine-condition multi-seed ablation for the larger architecture. The observed effects—including the task-dependent performance asymmetry—may be sensitive to model family, parameter count, or inference backend beyond the two evaluated scales.

Single-split evaluation.

Results are based on a single scaffold-split test set for each dataset (BBBP $N = 204$, BACE $N = 152$) without cross-validation. This makes the reported ROC-AUC values highly sensitive to the specific test partition. Given the small test set sizes, the observed differences between conditions (e.g., the 0.093 gap between C0 and C4a on BBBP) could be artifacts of the specific split rather than robust generalizable phenomena.

Limited sample size.

With 204 and 152 molecules respectively, this study is underpowered for detecting small but potentially meaningful effect size differences between conditions. The SP category comprised only 5 naturally occurring instances in BBBP, making any distributional claims about SP highly tentative.

C2 control confound.

The chemical priming condition (C2) contains domain-relevant vocabulary ("structural", "molecular", "physicochemical"), making it a partial rather than pure stylistic control. While C2b addresses this with non-chemical vocabulary, the C2 results should be interpreted with this confound in mind.

Heuristic classifier validity.

The heuristic classification scheme used to categorize free hallucinations (C3) into SP, PI, MF, or CC is an unvalidated rule-based system. It lacks inter-rater reliability testing against human expert annotation, and its strict keyword matching may result in high false-negative rates for semantic variations. The Contextual Confabulation (CC) category functions as a catch-all residual class, encompassing heterogeneous subtypes whose homogeneous treatment obscures important variations. This limitation is further evidenced by the post-hoc verification of C4a–c outputs (Table 2): despite prompt-constrained forcing, C4a achieved 0% target compliance (SP), C4b achieved 25.5% (PI), and C4c achieved 20.4% (MF), with CC dominating all three conditions ($\geq$74%). This systematic prompt–output misalignment demonstrates that the classifier's narrow keyword sets fail to capture the full semantic range of generated descriptions, motivating future development of embedding-based or LLM-assisted taxonomy classifiers with broader semantic coverage.

Data leakage risk.

The Llama-3.1 pre-training corpus likely includes MoleculeNet datasets or related molecular property data. Consequently, the zero-shot predictions evaluated here may actually reflect varying degrees of retrieval rather than genuine inferential reasoning. Hallucinated text might inadvertently trigger memorized associations, collapsing the evaluation framework from a test of reasoning to a test of prompted recall.

API non-determinism.

Despite setting temperature = 0.0 and seed = 42, Groq's API documentation notes that seed-based reproducibility is provided on a best-effort basis. Exact numerical reproducibility across runs is therefore not guaranteed.

Dataset scope.

Only two MoleculeNet benchmarks were evaluated, representing physicochemical (BBBP) and enzymatic (BACE) prediction tasks. Generalization to other task types (e.g., toxicity, solubility, protein binding) remains untested.

Inference-only evaluation.

While we include a traditional ML baseline (Random Forest) for context, we evaluate only zero-shot LLM inference. Comparison with fine-tuned LLM models would provide additional context for the practical significance of the observed effects.

Methodological confounds.

The BACE Mechanism Fabrication (C4c) condition directly prompts the LLM to hypothesize biological targets. Because BACE-1 is the implied target of the BACE dataset, the MF text may inadvertently contain information about BACE-1 itself, creating a direct semantic confound rather than a general structural perturbation. Although we validated the C5 control across multiple permutation seeds (0.49 ± 0.02), a larger number of permutations with full-dataset evaluation would further strengthen this control.

7 Conclusion

The primary contribution of this work is a controlled semantic perturbation framework for systematically isolating semantic effects in LLM-based molecular prediction. Through a nine-condition ablation, we find that the type of structured perturbation—not merely its presence—is associated with differential outcomes in zero-shot prediction tasks. Prompt-constrained structural augmentation (C4a) yields the largest observed ROC-AUC increase on BBBP (0.626 vs. 0.533 baseline), though none of the BBBP shifts reached statistical significance ($p > 0.05$). Post-hoc verification reveals 0% SP classification for C4a, indicating that the effect reflects the prompt's structural topic focus rather than hallucination type—highlighting a systematic gap between prompt design intent and output taxonomy. On BACE, most hallucination conditions produce significant performance degradation (e.g., C3: 0.504 vs. 0.674 baseline, $p = 0.016$), with the notable exception of C4a (0.666, $p = 0.808$), which largely preserves baseline discriminative capacity.

Our random-permutation semantic control (C5) provides evidence that the observed directional shifts on BBBP are not attributable solely to scientific writing style, as disrupting molecule-description alignment eliminates discriminative improvement while preserving linguistic register.

These findings should be interpreted within the scope of this exploratory study: two LLM scales, two datasets, and single-split evaluation. This work provides a structured empirical framework to guide future investigation into when and why semantic perturbations affect LLM-based molecular reasoning.

Future directions.

Key extensions include: (1) investigation of the distributional compression phenomenon on additional enzymatic datasets, (2) expansion of multi-model validation across diverse LLM architectures and scales, (3) larger-scale evaluation with cross-validation, and (4) comparison with fine-tuned LLM models and additional traditional ML methods to further contextualize the practical significance of the observed effects.

Declarations.

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Data Availability. The complete codebase, generated datasets, and high-resolution figures are publicly available at: https://github.com/apiprdt/AI-Hallucinations.

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Authors' contributions. **Muhammad Afif Erdita:** Conceptualization, Data Curation, Formal Analysis, Investigation, Methodology, Project Administration, Software, Validation, Visualization, Writing – Original Draft, Writing – Review & Editing.

Use of AI and AI-assisted technologies. During the preparation of this work, the author used Llama-3.1-8B and Llama-3.3-70B for the generation of experimental hallucinations and molecular property prediction. These tools were also used for linguistic refinement and drafting of the manuscript. The author takes full responsibility for the content of the publication.

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A Appendix

A.1 Prompt Templates

All prediction tasks used the following zero-shot prompt structure with temperature = 0.0, `max_tokens` = 10, and seed = 42:

```
Given this molecule (SMILES: {smiles})
[Additional information: {augmentation}]

On a scale of 0 to 1, what is the probability
that this molecule can {task_description}?
Respond with ONLY a number between 0 and 1.
Nothing else.
```

The augmentation line is omitted for C0. Task descriptions: "penetrate the blood-brain barrier" (BBBP), "inhibit the BACE-1 enzyme" (BACE).

Hallucination generation used temperature = 0.9, `max_tokens` = 150, with the following type-specific prompts:

C3 (Free Hallucination):

"You are a speculative chemist. Given SMILES: {smiles}. Generate a creative, scientifically-sounding description of this molecule's potential properties and biological mechanisms. Be creative and speculative. Write 2–3 sentences only."

C4a (Structural Augmentation):

"Given SMILES: {smiles}. Describe the structural features, functional groups, and atomic composition. Be specific and speculative."

C4b (Property Inversion):

"Given SMILES: {smiles}. Describe its solubility, lipophilicity, and membrane permeability with specific values."

C4c (Mechanism Fabrication):

"Given SMILES: {smiles}. Hypothesize biological targets and cellular mechanisms. Be specific about protein targets."

A.2 Control Condition Templates

C2 (Chemical Priming):

Template-generated text using scientific vocabulary: *"This compound exhibits {adj} structural coherence with {noun} interaction domains, demonstrating {adj} physicochemical characteristics under {noun} conditions."*

Adjective pool: moderate, intermediate, variable, standard, conventional, typical. Noun pool: hypothetical, theoretical, computational, structural, molecular, chemical.

Note: This template contains chemical vocabulary ("structural", "molecular", "physicochemical"), making C2 a *partial* stylistic control. See C2b for the stricter control.

C2b (Pure Gibberish):

Template-generated text using exclusively non-chemical vocabulary: *"This entity exhibits {adj} portfolio coherence with {noun} governance domains, demonstrating {adj} bureaucratic characteristics under {noun} conditions."*

Adjective pool: variable, standard, conventional, typical, moderate, intermediate. Noun pool: portfolio, transactional, governance, bureaucratic, administrative, regulatory.

Vocabulary audit confirmed 0% overlap with a curated chemical/biological term list.

A.3 Taxonomy Classifier Pseudocode

1. **SP Detection:** For each element in {F, Cl, Br, I, P, S, N}, check if associated keywords appear in the generated text. Cross-reference with `mol.GetAtoms()` to verify element absence. If a keyword matches an absent element, classify as SP.
2. **PI Detection:** Compute MolLogP via RDKit. If LogP > 3 and text contains "water soluble", "hydrophilic", or "highly soluble", classify as PI. If LogP < 1 and text contains "lipophilic", "hydrophobic", or "membrane permeable", classify as PI.
3. **MF Detection:** Keyword matching for 9 biological targets: ACE2, HER2, EGFR, VEGF, p53, BCL-2, COX-2, dopamine, serotonin.
4. **CC (Default):** If none of the above conditions trigger, classify as Contextual Confabulation.

A.4 Representative Hallucination Examples

- **SP Example (SMILES: c1ccccc1, Benzene):** "Contains a reactive chlorine atom that enhances metabolic stability." (Benzene contains no chlorine atoms.)
- **PI Example (SMILES: high-LogP compound):** "This highly water-soluble molecule readily dissolves in aqueous media." (Contradicts computed LogP > 3.)
- **MF Example (Diazepam):** "Acts as a potent inhibitor of the VEGF protein kinase receptor." (Diazepam is a GABA receptor modulator, not a VEGF inhibitor.)
- **CC Example:** "This molecule shows promising drug-like properties and may have therapeutic applications in treating various diseases." (Generic, non-falsifiable.)

A.5 API Failure Rates

Across the BBBP experiment (204 molecules $\times$ 9 conditions = 1,836 predictions), API failures (rate limits, parsing errors) that defaulted to $p = 0.5$ accounted for less than 2% of total predictions. The exact count is logged in the experiment checkpoint files.

A.6 Implementation Environment

Experiments were executed on a local system (Windows 11) using the Groq API for LLM inference. Total runtime for the pilot study was approximately 4.5 hours. Library versions: `rdkit-pypi` v2023.9.1, `deepchem` v2.7.1, `groq` Python SDK, `scipy` v1.11, `scikit-learn` v1.3.